What did the framework actually improve?

Numerical question
Did the action satisfy the stated electrical limits?

Workflow question
Was the right study run and attached to this action?

Operator question
Can a reviewer identify the action and its remaining uncertainty?

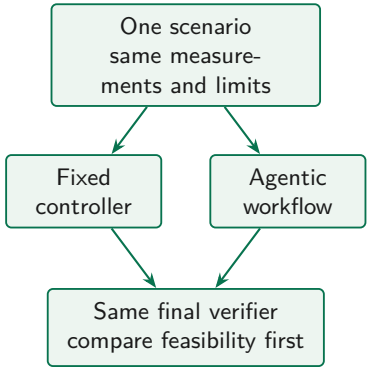
Measure the workflow and the electrical outcome separately.

- Study completed?
- Constraints passed?
- Claims supported?

Predict: Which claim is supported by a low cost and a fluent explanation?

PowerAgentBench-SS (2026), §§II–III; Executed tutorial: Classroom_Labs, L0–L12

Compare methods on the same task



Give every method the same physical constraints and verification budget.

For case i, record whether every constraint passes.

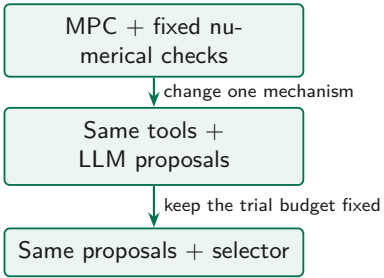
$$\widehat{p} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[\text{all checks pass}]$$

N is the number of cases. If 8 of 10 pass, $\widehat{p} = 0.8$. This arithmetic example is illustrative.

- Same case and admissible actions
- Same candidate trial budget
- Same final verifier

PowerAgentBench-SS (2026), §§II–III; Recorded voltage pilot: E1–E8

Ablate one mechanism at a time

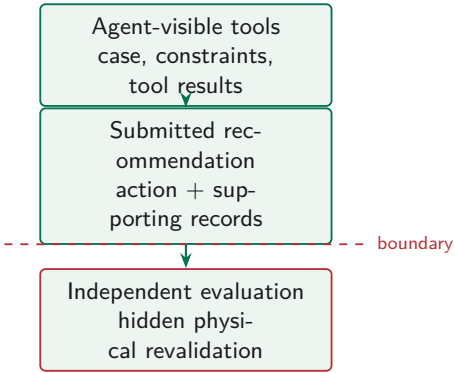


A useful ablation keeps the numerical tools fixed.

- Fixed numerical workflow
- Add language interpretation
- Add role-specific handoffs

PowerAgentBench-SS (2026), §§II–III; Executed LangGraph teaching lab: G0–G5; Recorded voltage pilot: E1–E8

Frontier: evidence-backed evaluation

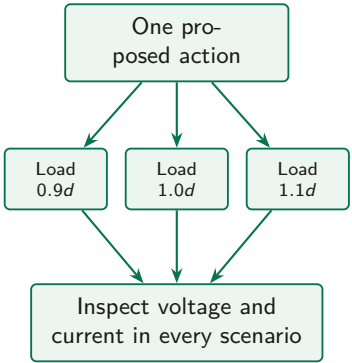


The final report should survive independent recomputation.

- ✓Public tools and fixed budgets
- ✓Hidden physical revalidation
- ✓Evidence-supported submissions

PowerAgentBench-SS (2026), §§II–III

Frontier: request uncertainty studies



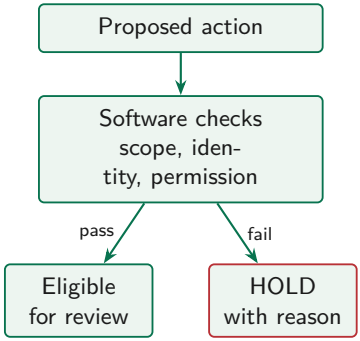
The orchestrator can request a stronger study when uncertainty matters.

Illustrative sensitivity study: use base demand d and two nearby load levels. Keep the same action in all three trials.

- ✓Numerical engines remain explicit
- ✓Uncertainty-aware study selection
- ✓Corrective-dispatch tools

CONDUCTOR (2026), uncertainty-aware orchestration; Executed tutorial: Classroom_Labs, L0–L12

Frontier: enforce authority in software



Approval rules belong in the runtime, not only in the prompt.

- ✓Whitelisted tools
- ✓Explicit step budgets
- ✓Typed numerical facts

Governance-aware TSO digital twin (2026); Executed tutorial: Classroom_Labs, L0–L12

L11: one optional local Llama call

Notebook L11 Exact notebook excerpt; run the complete cell in order.

```
client = ProposalLlama(journal, model='llama3.1:8b')
live_reply = client.ask('Optimization Agent', pool['context']['public_context'],
                        list(pool['context']['catalog']))
```

Actual saved public response from Llama 3.1:8b

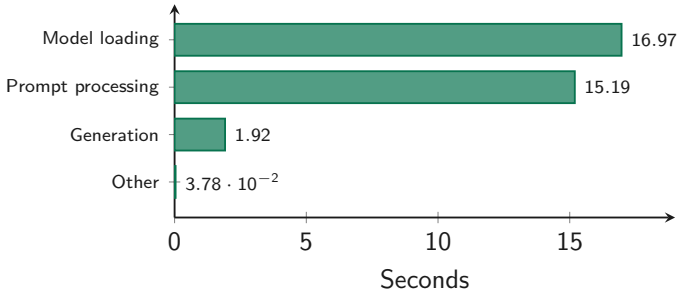
"has the potential to reduce reactive effort magnitude by ten percent, which may improve the cost."

Interpretation: this proposes an allowed trial. The new numerical check decides whether the change is feasible and useful.

Enable `RUN_LOCAL_LLAMA=True` for a new call. The saved response remains available if the local service is slow.

Ollama chat API; Classroom L11 and saved actual validation

Where did one Llama call spend its time?



Actual local Llama call
Client elapsed time: **34.113 s**.

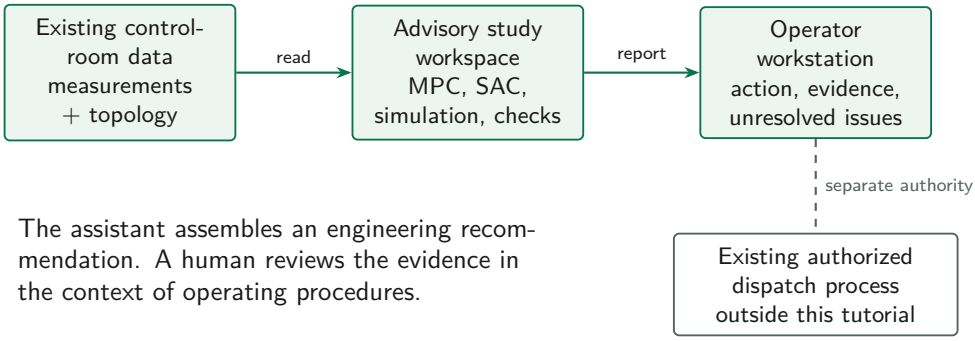
1,891 prompt tokens; 137 generated tokens.

Most time came before token generation. A warm repeat can behave differently.

Do not call a replay "inference latency".

Ollama chat API: timing fields; Executed tutorial: Classroom_Labs, L0–L12

How the workflow could fit beside control-room tools



The assistant assembles an engineering recommendation. A human reviews the evidence in the context of operating procedures.

The tutorial is an advisory sandbox with an explicit human boundary.

Governance-aware TSO digital twin (2026); Executed tutorial: Classroom_Labs, L0–L12

The operator sees scope before readback

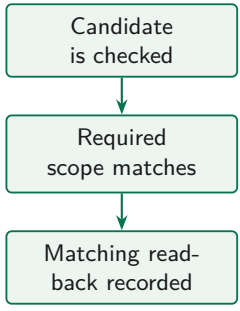
Saved recommendation ID: `b67cff9c8eb444c2`

Packet field	Full MV/LV request	Explicit MV teaching request
Verified scope	MV phases only	MV phases only
Scope matches	No	Yes
State	HOLD	READY_FOR_READBACK
Acknowledged	False	True
Equipment dispatch	False	False

Predict: The numerical evidence is the same. Why is only the second packet eligible?

Explanation: The second request asks only for the scope that was checked. The first asks for more. Acknowledgment still does not dispatch equipment.

Acknowledgment is a guarded transition



Only a matching, eligible packet can be acknowledged.

- Proposed → checked
- Checked → awaiting review
- Matching readback → acknowledged

Executed tutorial: Classroom_Labs, L0–L12; Executed LangGraph teaching lab: G0–G5; LangGraph: interrupts and resume

Code: bind review to the candidate

Notebook L12 Exact notebook excerpt; run the complete cell in order.

```
wrong_id = lab.acknowledge(mv_scope, 'wrong-id', 'Student operator')
readback = lab.acknowledge(mv_scope, mv_scope['recommendation_id'], 'Student operator')
```

What the code means

A wrong candidate ID cannot acknowledge the packet.

- First request uses a wrong ID
- Second reads back the packet ID
- The operator label is a classroom placeholder

Executed tutorial: Classroom_Labs, L0–L12

Hands-on: read back, reject, inspect

Four-minute notebook exercise

Use L12 to distinguish scope, identity and acknowledgment.

1. Run L12 after its earlier cells
2. Find the full-scope HOLD
3. Inspect the wrong-ID rejection
4. Confirm no equipment dispatch

Predict: Which of the three outcomes is permission to dispatch equipment?

Explanation: None. Even the acknowledged packet is explicitly advisory.

Executed tutorial: Classroom_Labs, L0–L12

The checks distinguish three outcomes

Notebook L12 Exact notebook excerpt; run the complete cell in order.

```
assert full_scope['state'] == 'HOLD' and not full_scope['acknowledged']
assert wrong_id['acknowledgment_result'] == 'REJECTED'
assert readback['acknowledged'] and not readback['equipment_dispatch']
```

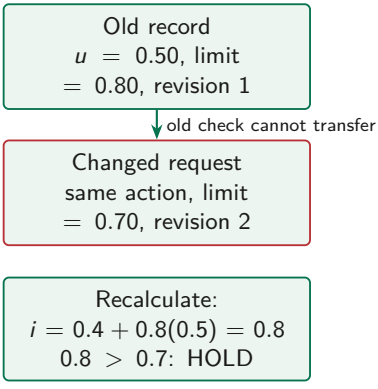
What the code means

Scope mismatch holds; wrong identity rejects; valid readback remains advisory.

- Full MV/LV requirement: HOLD
- Wrong identifier: REJECTED
- Matching MV readback: acknowledged

Executed tutorial: Classroom_Labs, L0–L12

A smaller current limit changes the decision



A hash links content; it does not establish physical truth.

u is the proposed control; i is the toy current. This uses the opening scalar model, not a feeder calculation.

- Change a limit or candidate
- Old binding no longer matches
- Recompute the affected study

Predict: Can the old acknowledgment survive a changed control?

Executed tutorial: Classroom_Labs, L0–L12

Capstone: approve review, hold or request a study?

Work with a partner

Name the missing evidence and the next responsible step.

1. A: MV PASS; full-feeder AC FAIL
2. B: restoration PASS; isolation record absent
3. C: no alarm; protected measurements absent

Predict: For your card, identify one missing fact and the next tool or human step.

Explanation: A: broader-model repair. B: authorized isolation evidence. C: justified independent measurements or escalation.

Executed tutorial: Classroom_Labs, L0–L12; Liu, Ning & Reiter, false-data injection, CCS 2009

Five stages, three cases, one evidence discipline

Stage	Voltage	Restoration	Cyber response
Awareness	Operating point and limits	Fault, ties, load priorities	Measurements and trust assumptions
Planning	MPC; SAC candidate	Weighted pickup MILP	Constrained recovery QP
Simulation	Network response	Flow on selected paths	Estimated / evaluated flows
Verification	Voltage/current and scope	Capacity, connectivity, binding	Residual and protected-data basis
LLM Optimizer	Propose a permitted refinement	Explain or request a revised trial	Explain or request additional evidence

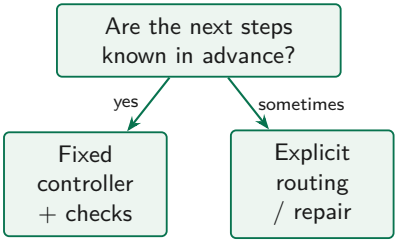
The numerical method changes with the decision. The record passed between stages keeps the action, assumptions and results together.

Predict: Which stage would catch a stale network revision?

Explanation: The evidence-binding check in Verification; a fluent summary cannot repair the mismatch.

Executed tutorial: Classroom_Labs, L0–L12; Recorded voltage pilot: E1–E8

Choose the simplest sufficient workflow



Add language interpretation only where requests or evidence require it. Test whether it improves a measured outcome.

Add an agent only when it addresses a measured task need.

- Repeated calculation → fixed workflow
- Varied requests → typed LLM interface
- Independent studies → specialist coordination

Kim et al., scaling agent systems (2026); Executed LangGraph teaching lab: G0–G5

A reproducible result carries its context

Experiment record

Scenario: load, topology, revision
Candidate: control values + ID
Model: equations + network scope
Checks: margins and failed limits
Time: solver, LLM, full workflow
Decision: accept for review / HOLD

Archive the inputs, methods, outputs and evidence mode together.

- Case, model scope and data revision
- Solver, policy and LLM versions
- Seeds, budgets and measured durations

Executed tutorial: Classroom_Labs, L0–L12; Recorded voltage pilot: E1–E8; Ollama chat API: timing fields

Design your next experiment

Work with a partner

Change one factor and state what would falsify your claim.

1. Hypothesis: name the expected benefit
2. Control: retain a strong fixed baseline
3. Measure: outcome, failures and latency
4. Boundary: state the model and task scope

Predict: Write one testable benefit and one result that would refute it.

Explanation: Example: fewer unsupported reports at equal tool budget; equal or higher unsupported-report rate refutes the claimed benefit.

PowerAgentBench-SS (2026), §§II–III; Kim et al., scaling agent systems (2026)

Exit ticket: explain, run, interpret

Work with a partner

Connect one concept, one cell and one observed result.

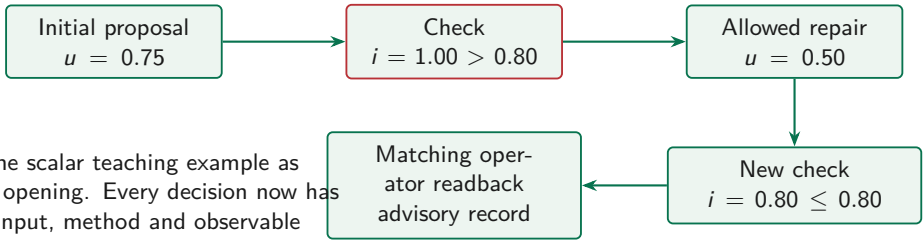
1. Explain a DAG versus a bounded cycle
2. Name a model-based and model-free method
3. Interpret one failed candidate correctly
4. Identify the operator acknowledgment boundary

Predict: Which checkpoint would you show a colleague to explain the framework?

Explanation: Choose a checkpoint and support the explanation with its exact inputs, output and stated limitation.

Executed LangGraph teaching lab: G0–G5; Executed tutorial: Classroom_Labs, L0–L12

From request to a reviewable recommendation



Same scalar teaching example as the opening. Every decision now has an input, method and observable output.

Agentic assistance earns trust through useful tools, explicit evidence and accountable decisions.

Executed tutorial: Classroom_Labs, L0–L12; Executed LangGraph teaching lab: G0–G5; Recorded voltage pilot: E1–E8